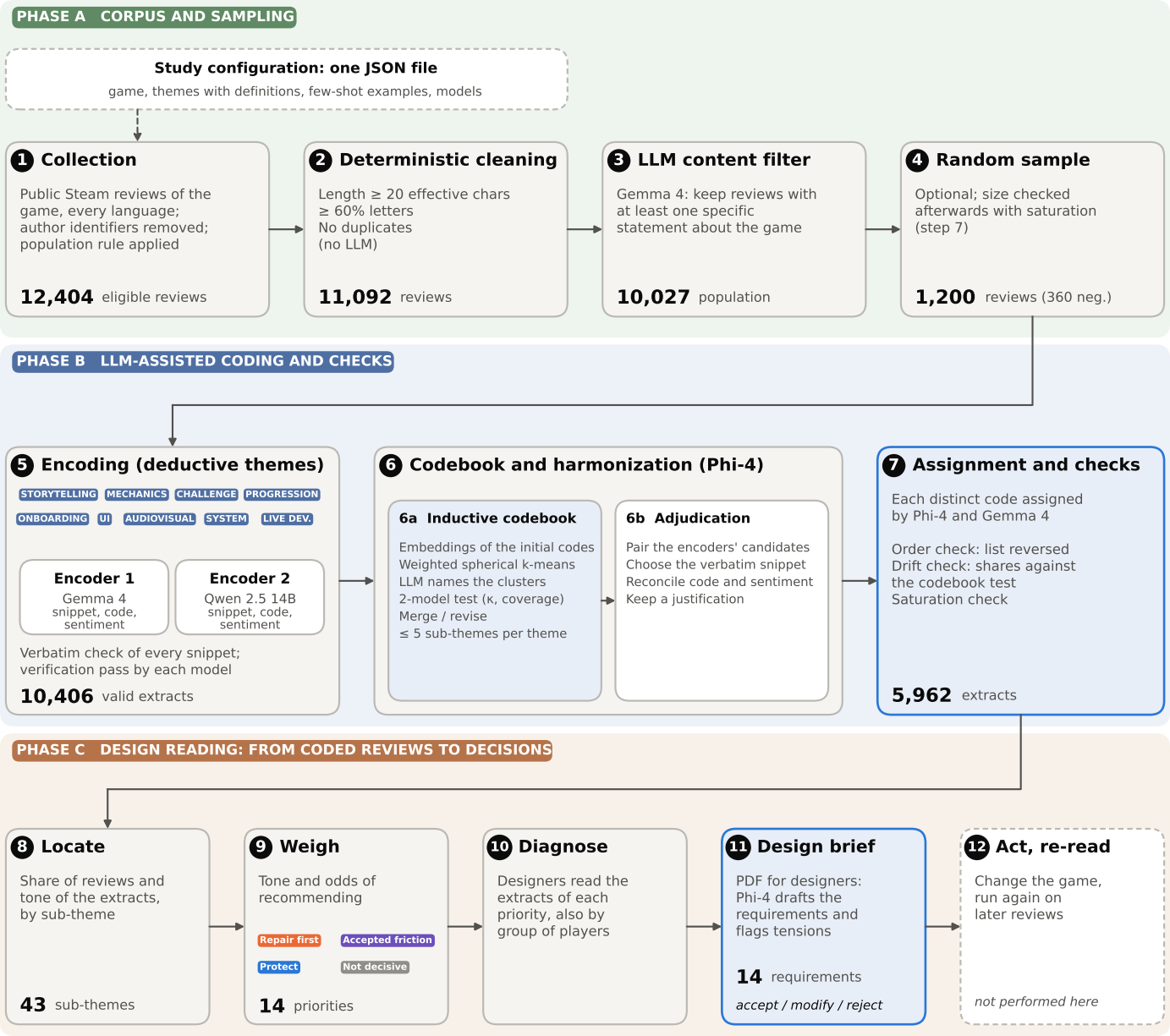
PHASE A CORPUS AND SAMPLING
Study configuration: one JSON file
game, themes with definitions, few-shot examples, models
1 Collection
Public Steam reviews of the game, every language; author identifiers removed; population rule applied
12,404 eligible reviews
2 Deterministic cleaning
Length ≥ 20 effective chars
≥ 60% letters
No duplicates
(no LLM)
11,092 reviews
3 LLM content filter
Gemma 4: keep reviews with at least one specific statement about the game
10,027 population
4 Random sample
Optional; size checked afterwards with saturation (step 7)
1,200 reviews (360 neg.)
PHASE B LLM-ASSISTED CODING AND CHECKS
5 Encoding (deductive themes)
STORYTELLING
MECHANICS
CHALLENGE
PROGRESSION
ONBOARDING
UI
AUDIOVISUAL
SYSTEM
LIVE DEV.
Encoder 1
Gemma 4
snippet, code, sentiment
Encoder 2
Qwen 2.5 14B
snippet, code, sentiment
Verbatim check of every snippet; verification pass by each model
10,406 valid extracts
6 Codebook and harmonization (Phi-4)
6a Inductive codebook
Embeddings of the initial codes
Weighted spherical k-means
LLM names the clusters
2-model test (κ, coverage)
Merge / revise
≤ 5 sub-themes per theme
6b Adjudication
Pair the encoders' candidates
Choose the verbatim snippet
Reconcile code and sentiment
Keep a justification
7 Assignment and checks
Each distinct code assigned by Phi-4 and Gemma 4
Order check: list reversed
Drift check: shares against the codebook test
Saturation check
5,962 extracts
PHASE C DESIGN READING: FROM CODED REVIEWS TO DECISIONS
8 Locate
Share of reviews and tone of the extracts, by sub-theme
43 sub-themes
9 Weigh
Tone and odds of recommending
Repair first
Accepted friction
Protect
Not decisive
14 priorities
10 Diagnose
Designers read the extracts of each priority, also by group of players
11 Design brief
PDF for designers: Phi-4 drafts the requirements and flags tensions
14 requirements
accept / modify / reject
12 Act, re-read
Change the game, run again on later reviews
not performed here